Abraham Jhared Flores Azcona

AI/ML and Systems Engineer

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Engineer with experience developing and evaluating applied AI prototypes, integrating software for resource-constrained devices, and conducting reproducible machine-learning experiments. Public work spans representation analysis, scientific ML, MIDI/SysEx systems, automated testing, and cross-platform tooling.

TECHNICAL SKILLS

Programming: Python, C, C++, TypeScript, JavaScript | ML and scientific computing: PyTorch, Hugging Face Transformers, scikit-learn, SciPy, NumPy, Pandas
Systems and development: Linux, Windows, Git/GitHub, Jupyter, Google Colab, LaTeX, SDK/API integration, MIDI/SysEx, CI testing
Methods: reproducible experimentation, representation analysis, activation steering, PCA, cosine similarity, ROC analysis
Languages: Spanish (native); English (TOEFL iBT 5.5/6.0)

EXPERIENCE

Assistant Engineer - Open Innovation | Samsung Research Tijuana
Feb 2024 - Present

- Developed and evaluated 10+ AI proof-of-concept prototypes for resource-constrained consumer-device environments, including on-device ML, language, speech, and assistant-related capabilities.
- Integrated and tested partner SDKs and APIs using C/C++, Python, TypeScript, and JavaScript; assessed feasibility, platform constraints, and engineering trade-offs while respecting product confidentiality.
- Collaborated with international engineering teams and technology partners; delivered four technical demonstrations to executive stakeholders during workshops in South Korea.

Undergraduate Thesis Researcher | Instituto Tecnológico de Tijuana
Aug 2023 - Jan 2024
Advisor: José Sergio Magdaleno Palencia | Thesis submitted Jan 2024; professional examination passed Oct 7, 2024.

- Co-designed a quasi-experimental pretest-posttest study involving 46 consenting engineering students and analyzed outcomes for 39 participants across control and digitally assisted groups.
- Built a Python analysis pipeline using Pandas, NumPy, SciPy, Matplotlib, and NLTK; reported descriptive findings, confounders, and methodological limitations and coauthored the thesis.

Research Assistant - ENLACE 2023 Summer Research Program | University of California San Diego
Jun 2023 - Aug 2023
Mentor: Jonathan Guiang

- Implemented preprocessing, training, inference, and ROC evaluation in PyTorch for CMS candidate line-segment classification on a UC San Diego computing cluster.
- Benchmarked two fully connected classifiers against a laboratory GNN baseline; the larger DNN achieved ROC-AUC 0.9724 versus 0.9751 for the GNN.
- Coauthored the code and technical report and presented the work at UC San Diego and Instituto Tecnológico de Tijuana symposia.

Academic Tutor | Universidad Autónoma de Baja California
Mar 2023 - Oct 2023
Supervisor: Lizeth Carolina Aguilar Dodier

- Tutored 62 undergraduate engineering students in Differential Calculus, Programming Methodology, and Programming and Numerical Methods through individual and group sessions.
- Adapted explanations and exercises to students' preparation and coordinated topic-focused study groups.

SELECTED PUBLIC PROJECTS

Affective Representation Analysis and Activation Steering | Independent Researcher | Repository
Apr 2026 - Present

- Conducted a resource-constrained partial replication of selected emotion-concept analyses across GPT-2 Medium and Gemma 4 E2B using 9- and 20-emotion configurations.
- Released four reproducible Colab configurations and analyzed representations from 2,000 generated stories using Logit Lens projections, PCA, cosine similarity, and activation interventions.
- Documented model-, prompt-, corpus-, control-, and evaluation-related limitations and avoided claims beyond the observed configurations.

BlueBoard - BOSS Katana CLI Bridge | Independent Developer | v1.0.0 repository
2026

- Released a Python CLI that maps iRig BlueBoard BLE-MIDI input to BOSS Katana amplifier controls and configurable desktop actions.
- Implemented MIDI event routing, reconnection handling, modular action dispatch, and SysEx state synchronization for the original Katana MkI.
- Maintained 167 automated tests with Windows/Linux CI and package smoke testing; physically demonstrated the v1.0.0 release on Windows.

EDUCATION

Ingeniero en Sistemas Computacionales (Computer Systems Engineering) | Instituto Tecnológico de Tijuana, Tecnológico Nacional de México
Degree conferred Nov 2024 | Coursework completed Dec 2023 | GPA: 95.12/100 | 260/260 credits
Selected coursework: Linear Algebra; Vector Calculus; Differential Equations; Probability and Statistics; Discrete Mathematics; Numerical Methods; Data Structures; Computer Architecture; Operating Systems; Formal Languages and Automata; Artificial Intelligence; Operations Research; Simulation.

SELECTED PRESENTATIONS

Affective Representation Analysis and Activation Steering in GPT-2 Medium and Gemma 4 E2B. Internal research seminar, Samsung Research Tijuana, May 12, 2026.
Tizen Native in the Age of AI Tools. INNOVATEC 2025, Instituto Tecnológico de Tijuana, Nov 20, 2025.
Using Machine Learning for Particle Tracking at the Large Hadron Collider. Oral presentation, ENLACE 2023 Simposio en Baja California, Sep 28, 2023; poster, UC San Diego ENLACE Research Symposium, Aug 11, 2023.

ADDITIONAL TRAINING

IBM Data Science Professional Certificate, Coursera, Jan 2026 | McKinsey.org Forward Program, McKinsey & Company, Dec 2025